Advanced Mathematical Techniques
Bookwork 3

1. **Problem 5.9.13:** For "small" values of x,

$$ln(x!) = -\gamma x + \sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)}{n} x^n$$

where γ is the Euler-Mascheroni constant and $\zeta(n)$ is the Riemann zeta function. For what values of x does this series converge?

By the Ratio Test:

$$L = \lim_{n\to\infty} \left| \frac{(-1)^{n+1}\zeta(n+1)x^{n+1}}{n+1} \cdot \frac{n}{(-1)^n \zeta(n) x^n} \right| = \lim_{n\to\infty} |x| < 1$$

Therefore $-1 < x < 1$. Checking the endpoints,

$$\sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)}{n} (1)^n = \sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)}{n}$$

The above is a variation on the alternating harmonic series which converges. Therefore 1 works.

$$\sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)}{n} (-1)^n = \sum_{n=2}^{\infty} (1)^n \frac{\zeta(n)}{n}$$

The above is a variation on the harmonic series which diverges. Therefore -1 doesn't work.
Thus:

$$\boxed{-1 < x \leq 1}$$

2. **Problem 8.1.7:** Show that:

$$\int_0^{\infty} e^{-x^4} dx = \Gamma\left(\frac{5}{4}\right)$$

We derived in class that:

$$\int_0^{\infty} x^{m-1} e^{-ax^n} dx = \frac{1}{na^{\frac{m}{n}}} \Gamma\left(\frac{m}{n}\right)$$

Plugging in,

$$\int_0^{\infty} x^{1-1} e^{-1\cdot x^4} = \frac{1}{4\cdot(1)^{\frac{1}{4}}} \Gamma\left(\frac{1}{4}\right) = \frac{1}{4}\Gamma\left(\frac{1}{4}\right) = \Gamma\left(\frac{5}{4}\right) \checkmark$$

3. **Problem 8.1.8:** Show that:

$$\lim_{x\to 0} \frac{(ax-1)!}{(x-1)!} = \frac{1}{a}$$

$$\lim_{x\to 0} \frac{(ax-1)!}{(x-1)!} = \lim_{x\to 0} \frac{\Gamma(ax)}{\Gamma(x)} = \lim_{x\to 0} \frac{\frac{\Gamma(ax+1)}{ax}}{\frac{\Gamma(x+1)}{\Gamma(x)}} = \lim_{x\to 0} \left(\frac{1}{a} \cdot \frac{\Gamma(ax+1)}{\Gamma(x+1)} \right) = \frac{1}{a} \cdot 1 = \boxed{\frac{1}{a}}$$